COURSE CONTENT SUMMARY — FOR TRANSFER EVALUATION

CHEM&131: Introduction to Organic and Biochemistry

Fall 2026 (September 21 – December 10, 2026)

What's Covered

Everything covered in this course, organized by topic and week — useful for comparing against an equivalent course at another institution, and for previewing what's ahead before the term starts.

Built from the Fall 2026 Chemfolio schedule — first term using this material, so some groupings may shift slightly as the term settles in.

Week 1: Introduction to Organic Molecules and Alkanes

What this week covers:

- Characteristic features of organic compounds; drawing organic molecules
- Functional groups; alkanes and alkane nomenclature; cycloalkanes
- Fossil fuels, physical properties, and combustion of hydrocarbons
- Lab: Safety orientation; scavenger hunt covering compounds, shapes, and IMFs
- Required Reading: Smith Ch. 10 (10.1–10.10)
- Naming guide: How a Name Is Built, Alkanes and Cycloalkanes
- Structure guide for lab workbook Appendices 1–3: Molecular Shapes, Molecular Polarity, Solubility of Organic Compounds

▼ **Weekly logistics**

- Activity: A21 (10.1–10.6)
- Activity: A22 (10.4–10.10)

Week 2: Alkenes, Alkynes, Aromatics, and Isomers

What this week covers:

- Alkenes and alkynes; their nomenclature
- Aromatic compounds; nomenclature of benzene derivatives; sunscreens and antioxidants
- Constitutional and cis-trans (geometric) isomers; alkane/cycloalkane naming revisited
- Lab: Solubility and Oil Spills
- Required Reading: Smith Ch. 11 (11.1–11.2, 11.3–11.4, 11.8–11.10), Ch. 10 (10.5–10.8)
- Naming guide: Alkenes, Alkynes, and Cis/Trans, Aromatic Compounds

▼ **Weekly logistics**

- Activity: A23 (11.1–11.2, 11.8–11.10)
- Activity: A24 (10.5–10.8, 11.3–11.4)
- Writing Workshop 1

Week 3: Alkene Reactions, Polymers, and Alcohols

What this week covers:

- Reactions of alkenes; polymers; margarine vs. butter
- Alcohols and ethers — structure, properties, and reactions; alkyl halides; thiols
- Lab: Plastics and Nurdles
- Library Visit 1
- Required Reading: Smith Ch. 11 (11.5–11.7), Ch. 12 (12.1–12.7)
- Naming guide: Alcohols, Ethers, Halides, Nitro Compounds, and Thiols
- Reactions guide: Identify the Reaction Type, Addition and Elimination, Oxidizing Alcohols

▼ **Weekly logistics**

- Activity: A25 (11.5–11.7)
- Activity: A26 (12.1–12.7)

Week 4: Oxygenated Compounds and Carboxylic Acids

What this week covers:

- Aldehydes and ketones; oxidation of aldehydes; chirality (Looking Glass Chemistry); chiral drugs
- Carboxylic acids and esters — naming, properties, acidity, and reactions
- Lab: Chirality
- Required Reading: Smith Ch. 12 (12.8–12.12), Ch. 13 (13.1–13.6)
- Naming guide: Aldehydes and Ketones, Chirality, R and S, and Meso Compounds, Carboxylic Acids, Esters, and Anhydrides
- Reactions guide: Oxidation and Reduction, Esters: Condensation and Hydrolysis, Carboxylic Acids Give Up H^+

▼ **Weekly logistics**

- Activity: A27 (12.8–12.12)
- Activity: A28 (13.1–13.6)
- Also this week: Unit Quiz 1 (Ch. 10 + Ch. 11)

Week 5: Amines, Amides, and Protein Structure

What this week covers:

- Amines and amides — structure, basicity, naming
- Amino acids, acid-base behavior of amino acids, and peptides
- Lab: Reactions of Functional Groups
- Library Visit 2
- Required Reading: Smith Ch. 13 (13.7–13.10), Ch. 16 (16.1–16.7)
- Naming guide: Amines and Amides

- Reactions guide: Amides and Peptide Bonds, Amines Pick Up H^+, Amino Acids as Zwitterions

▼ **Weekly logistics**

- Activity: A29 (13.7–13.10)
- Activity: A30 (16.1–16.7)
- Writing Workshop 2

Week 6: Proteins, Enzymes, and Carbohydrates

What this week covers:

- Common proteins; protein hydrolysis and denaturation; enzymes and how they work; using enzymes to diagnose and treat disease
- Monosaccharides and their cyclic forms
- Glycolysis: first look, alongside carbohydrates (energy accounting returns in Week 10)
- Lab: Protein Light
- Required Reading: Smith Ch. 16 (16.8–16.11), Ch. 14 (14.1–14.3), Ch. 18 (18.4)
- Naming guide: Fischer Projections and D/L
- Reactions guide: Protein Digestion (Hydrolysis), Reading an Enzyme's Name, Sugar Rings (Hemiacetals)

▼ **Weekly logistics**

- Activity: A31 (16.8–16.11)
- Activity: A32 (14.1–14.3, 18.4)
- Also this week: Unit Quiz 2 (organic reactions and functional groups); Archer Gallery Visit in place of a lab

Week 7: Carbohydrates and Lipids

What this week covers:

- Reactions of monosaccharides; disaccharides; polysaccharides; blood type
- Lipids — fatty acids, waxes, triacylglycerols, and their hydrolysis
- Lab: Sugar Spill and Enzymes (Day 1); Bioremediation (Day 2)
- Required Reading: Smith Ch. 14 (14.4–14.7), Ch. 15 (15.1–15.5)
- Naming guide: Fatty Acids: Chain Length and Cis/Trans
- Reactions guide: Sugar Oxidation and Reduction, Reducing Sugars and Benedict's Test, Glycosidic Bonds and Triglycerides, Digestion and Saponification, Margarine and Trans Fats

▼ **Weekly logistics**

- Activity: A33 (14.4–14.7)
- Activity: A34 (15.1–15.5)
- Bioremediation lab finishes with a Graphing Workshop

Week 8: Cell Membranes and Nucleic Acids

What this week covers:

- Phospholipids and cell membranes; membrane transport and diffusion; cholesterol; steroid hormones; fat-soluble vitamins

- β-oxidation of fatty acids: first look, alongside lipids (energy accounting returns in Week 10)
- Nucleosides, nucleotides, and nucleic acids; the DNA double helix; replication; RNA
- Lab: Bioplastics (Design Challenge)
- Required Reading: Smith Ch. 15 (15.6–15.10), Ch. 17 (17.1–17.5), Ch. 18 (18.9)
- Naming guide: Steroids, Nitrogen Rings

▼ **Weekly logistics**

- Activity: A35 (15.6–15.10, 18.9)
- Activity: A36 (17.1–17.5)
- Also this week: Unit Quiz 3 (Ch. 16 + Ch. 14)

Week 9: Protein Synthesis and Metabolism Overview

What this week covers:

- Transcription; the genetic code; translation; mutations and genetic disease; modern DNA technology; viruses
- Overview of metabolism; ATP and energy production; coenzymes in metabolism
- Lab: DNA Extraction
- Required Reading: Smith Ch. 17 (17.6–17.11), Ch. 18 (18.1–18.3)
- Naming guide: Phosphates and ATP
- Reactions guide: Reactions in Metabolism, NAD^+ and FAD as Coenzymes, Reading an Enzyme's Name

▼ **Weekly logistics**

- Activity: A37 (17.6–17.11)
- Activity: A38 (18.1–18.3)
- Seed-toxicology lab prep begins this week (advance germination time)

Week 10: Catabolism, Energy Expenditure, and Final Project

What this week covers:

- Revisiting glycolysis (Week 6) and β-oxidation (Week 8) to account for every NADH, $FADH_2$, ATP, and GTP produced
- The fate of pyruvate; the citric acid cycle; electron transport chain and oxidative phosphorylation; ATP yield from glucose
- Ketone bodies; amino acid metabolism and the urea cycle
- Lab: Seed Toxicology
- Final Project: Teach Your Community (TYC) — final exam assessment, 10% of course grade, due Thursday, 12/10
- Required Reading: Smith Ch. 18 (18.4–18.11; 18.4 and 18.9 revisited from Weeks 6 and 8)
- Reactions guide: One Pattern, Used Again and Again, Combustion: The Whole Trip at Once

▼ **Weekly logistics**

- Activity: A39 (18.4–18.8; revisits 18.4)
- Activity: A40 (18.8–18.11; revisits 18.9)
- Also this week: Unit Quiz 4 (Ch. 15 + Ch. 17); Unit Quiz 5 (Ch. 18); TYC Outline work continues
- Finals week: course potluck